

DM–DE and DR–DE couplings in **darkin-class**: equations to implement

Transcribed from 2505.10410v2 + blackboard (12/06/2026)
matter sector cross-checked against 2212.08098 (Eqs. 15–17) and 1707.07667 (Eqs. 2.18–2.19)

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Conventions: $(-, +, +, +)$, $M_p = 1$, conformal time τ , $\mathcal{H} = a'/a$, $\dot{\phi} \equiv d\phi/dt$ (cosmic time), $\phi' \equiv d\phi/d\tau = a\dot{\phi}$. Throughout, $A'_m \equiv dA_m/d\phi$, $A'_r \equiv dA_r/d\phi$ (derivatives w.r.t. the field). Synchronous gauge for perturbations, CLASS/Ma–Bertschinger notation.

1 Model (paper 2505.10410, Sec. 2–3)

Action and Lagrangian (first blackboard line, cleaned):

$$S = \int d^4x \sqrt{|g_4|} \left(\frac{M_p^2}{2} R_4 - \mathcal{L} \right), \quad (1)$$

$$\mathcal{L} = \frac{1}{2} \partial_\mu \phi \partial^\mu \phi + V(\phi) + A_m(\phi) \bar{\mathcal{L}}_m + A_r(\phi) \bar{\mathcal{L}}_r, \quad \bar{\mathcal{L}}_r = \frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \text{gauge}. \quad (2)$$

On-shell identifications: $\bar{\mathcal{L}}_m = \bar{\rho}_m$, $\delta \bar{\mathcal{L}}_m = \delta \bar{\rho}_m$, and for the radiation (effective fluid) sector $\bar{\mathcal{L}}_r = \bar{\rho}_r$, $\delta \bar{\mathcal{L}}_r = \delta \bar{\rho}_r$.

Coupled densities [paper (2.19)]:

$$\rho_m = A_m(\phi) \bar{\rho}_m, \quad \bar{\rho}_m = \rho_{m0} a^{-3}, \quad \rho_r = A_r(\phi) \bar{\rho}_r, \quad \bar{\rho}_r = \rho_{r0} a^{-4}, \quad p_r = \frac{1}{3} \rho_r. \quad (3)$$

Coupling functions, **Eq. (3.11)** of the paper, same form for both sectors:

$$\boxed{A_m(\phi) = 1 + A_m^\phi \left(e^{\alpha_m \phi} - 1 \right), \quad A_r(\phi) = 1 + A_r^\phi \left(e^{\alpha_r \phi} - 1 \right).} \quad (4)$$

String-inspired values (3.12): $\lambda = \sqrt{8/3}$, $\alpha_m = \sqrt{3/8}$, $\alpha_r = \sqrt{3/2}$, $A_m^\phi = A_r^\phi = 1/8$.

Scalar field equation [paper (2.22)], cosmic time:

$$\ddot{\phi} + 3H\dot{\phi} + V_{,\phi} + \bar{\rho}_m A'_m + \bar{\rho}_r A'_r = 0. \quad (5)$$

2 Modified background equations (derivation; matches board of 12/06 and paper Sec. 2)

Cosmic time, $H = \dot{a}/a$, $M_p = 1$. From varying the Lagrangian w.r.t. ϕ , with $\bar{\mathcal{L}}_m = \bar{\rho}_m$, $\bar{\mathcal{L}}_r = \bar{\rho}_r$ on-shell [paper (2.22)]:

$$\boxed{\ddot{\phi} + 3H\dot{\phi} + V' + \bar{\rho}_m A'_m + \bar{\rho}_r A'_r = 0} \quad (6)$$

With $\rho_\phi = \frac{1}{2} \dot{\phi}^2 + V$, $p_\phi = \frac{1}{2} \dot{\phi}^2 - V$, multiplying (??) by $\dot{\phi}$:

$$\dot{\rho}_\phi = \dot{\phi} \ddot{\phi} + V' \dot{\phi} = -3H(1 + w_\phi) \rho_\phi - \bar{\rho}_m A'_m \dot{\phi} - \bar{\rho}_r A'_r \dot{\phi}. \quad [\text{board line 2; paper (2.23)}] \quad (7)$$

Coupled fluids: since $\rho_m = A_m(\phi)\bar{\rho}_m$ with $\bar{\rho}_m \propto a^{-3}$ (and similarly ρ_r with a^{-4} , $w_r = 1/3$),

$$\boxed{\dot{\rho}_m = -3H\rho_m + (\partial_\phi \ln A_m)\dot{\phi}\rho_m} \quad \boxed{\dot{\rho}_r = -3H\left(1 + \frac{1}{3}\right)\rho_r + (\partial_\phi \ln A_r)\dot{\phi}\rho_r} \quad (8)$$

[board lines 4–5]. Effective dark energy, defined so that Friedmann equations keep their standard form in terms of the *uncoupled* ($\propto a^{-3}, a^{-4}$) densities [board right block; paper (2.20)–(2.21)]:

$$\begin{aligned} \rho_\phi + \rho_m + \rho_r &= \rho_{\text{DE}} + \bar{\rho}_m + \bar{\rho}_r, & \rho_{\text{DE}} &\equiv \rho_\phi + (A_m - 1)\bar{\rho}_m + (A_r - 1)\bar{\rho}_r, & (9) \\ w_{\text{DE}} \rho_{\text{DE}} &= w_\phi \rho_\phi + \frac{1}{3}(A_r - 1)\bar{\rho}_r. & & & (10) \end{aligned}$$

Consistency check: differentiating ρ_{DE} and using (??)–(??), the explicit coupling terms cancel,

$$\dot{\rho}_{\text{DE}} = -3H(1 + w_\phi)\rho_\phi - 3H(A_m - 1)\bar{\rho}_m - 4H(A_r - 1)\bar{\rho}_r = -3H(1 + w_{\text{DE}})\rho_{\text{DE}}, \quad (11)$$

where the last equality holds precisely with definition (??) [board line 3; paper (2.24)]. Note that $w_{\text{DE}} \neq w_\phi \rho_\phi / \rho_{\text{DE}}$ once $A_r \neq 1$: the radiation coupling adds the $\frac{1}{3}(A_r - 1)\bar{\rho}_r$ term, contrary to the matter-only case (2.7).

Code form (conformal time, CLASS units $\rho_{\text{class}} = \rho_{\text{phys}}/3$)

With $\phi' = a\dot{\phi}$, $\mathcal{H} = aH$:

$$\phi'' + 2\mathcal{H}\phi' + a^2(V_{,\phi} - Q_m - Q_r) = 0, \quad Q_m = -3\rho_{\text{qcdm}}\frac{A'_m}{A_m}, \quad Q_r = -3\rho_{\text{idr}}\frac{A'_r}{A_r}, \quad (12)$$

$$\frac{d\rho_{\text{qcdm}}}{d\ln a} = -3\rho_{\text{qcdm}} - \frac{1}{3}Q_m\frac{\phi'}{\mathcal{H}}, \quad \frac{d\rho_{\text{idr}}}{d\ln a} = -4\rho_{\text{idr}} - \frac{1}{3}Q_r\frac{\phi'}{\mathcal{H}}. \quad (13)$$

Q_m is the existing `Q_scf` in the pure conformal limit ($A'_m/A_m = \frac{1}{2}C_{,\phi}/C$); Q_r is new. Equivalently $\rho_{\text{idr}} = A_r(\phi)\bar{\rho}_{\text{idr}}$, $\bar{\rho}_{\text{idr}} \propto a^{-4}$ can be used algebraically instead of integrating. $H^2 = \sum_n \rho_{n,\text{class}}$ unchanged, with ρ_{qcdm} , ρ_{idr} the *coupled* densities.

3 Blackboard perturbation equations (corrected transcription)

(Subscripts on the board are m and r , not x ; second photo adds the momentum equation.)

$$\delta T^0_0 = \frac{1}{2}\dot{\phi}^2 \delta g^{00} + g^{00}\dot{\phi}\delta\dot{\phi} - V'\delta\phi - A_m\delta\bar{\mathcal{L}}_m - A_r\delta\bar{\mathcal{L}}_r - A'_m\delta\phi\bar{\rho}_m - A'_r\delta\phi\bar{\rho}_r, \quad (14)$$

$$\delta T^i_0 = g^{ij}\partial_0\phi\partial_j\delta\phi, \quad \delta\Sigma^i_j = 0, \quad (15)$$

$$\delta T^i_j = -\delta^i_j\delta g^{00}\frac{\dot{\phi}^2}{2} - \delta^i_j g^{00}\dot{\phi}\delta\dot{\phi} - \delta^i_j V'\delta\phi + \frac{1}{3}\delta^i_j (A_r\delta\bar{\mathcal{L}}_r + A'_r\delta\phi\bar{\rho}_r), \quad (16)$$

$$\nabla_\mu\delta T^\mu_0 = \nabla_0(\phi, V, m, r) - (A'_m\bar{\rho}_m + A'_r\bar{\rho}_r)\delta\dot{\phi} - (A''_m\bar{\rho}_m + A''_r\bar{\rho}_r)\dot{\phi}\delta\phi + \dot{\phi}\Delta_3\delta\phi, \quad (17)$$

$$\nabla_\mu\delta T^\mu_j = g^{00}\ddot{\phi}\partial_j\delta\phi + g^{00}\dot{\phi}\nabla_0\partial_j\delta\phi - \nabla_j\left(\frac{1}{2}\delta g^{00}\dot{\phi}^2 + g^{00}\dot{\phi}\delta\dot{\phi} - V'\delta\phi\right) + \frac{1}{3}A_r\nabla_j\delta\bar{\mathcal{L}}_r + \underbrace{\frac{1}{3}A'_r\bar{\rho}_r\partial_j\delta\phi}_{\text{only new term}}. \quad (18)$$

Hence for the coupled radiation fluid:

$$\delta\rho_r = A_r\delta\bar{\rho}_r + A'_r\delta\phi\bar{\rho}_r, \quad \delta p_r = \frac{1}{3}\delta\rho_r, \quad (\rho_r + p_r)\theta_r = A_r(\bar{\rho}_r + \bar{p}_r)\bar{\theta}_r \Rightarrow \theta_r = \bar{\theta}_r. \quad (19)$$

4 Mapping to darkin-class (DM sector)

The code implements the conformal coupling via $C(\phi)$ (currently $C = C_0 e^{2\beta\phi}$), with background KG $\phi'' + 2\mathcal{H}\phi' + a^2(V_{,\phi} - Q) = 0$ and qcdm continuity $\rho'_{\text{qcdm}} = -3\mathcal{H}\rho_{\text{qcdm}} - \frac{1}{3}Q\phi'$; for pure conformal $Q = -\frac{3}{2}\rho_{\text{qcdm}}C'/C$ (CLASS units $\rho_{\text{class}} = \rho_{\text{phys}}/3$, hence the relative factor 3 between KG and continuity). This integrates to $\rho_{\text{qcdm}} \propto C^{1/2}a^{-3}$, so the dictionary is

$$\boxed{A_m = C^{1/2} \iff C(\phi) = A_m(\phi)^2 = \left[1 + A_m^\phi (e^{\alpha_m\phi} - 1)\right]^2.} \quad (20)$$

This is consistent with 1707.07667, Eqs. (2.18)–(2.19): $C = e^{2\alpha\kappa\phi}$ and $Q^{(c)} = -\frac{1}{2}(\ln C)_{,\phi}(1 - 3w_c)\rho_c$, so that the coupling strength is

$$\Lambda_m \equiv \frac{d \ln A_m}{d\phi} = \frac{A'_m}{A_m} = \frac{1}{2}(\ln C)_{,\phi}, \quad \Lambda_{2,m} \equiv \frac{d^2 \ln A_m}{d\phi^2} = \frac{A''_m}{A_m} - \Lambda_m^2. \quad (21)$$

Note the $(1 - 3w)$ factor in $Q^{(c)}$: the standard conformal (trace) coupling *vanishes* for radiation, which is precisely why the DR–DE coupling below is a different structure (multiplicative $A_r(\phi)\bar{\mathcal{L}}_r$) and needs new code.

Implementation: add a new selectable functional form for C_{scf} , dC_{scf} , ddC_{scf} :

$$C = A_m^2, \quad \frac{dC}{d\phi} = 2 A_m A'_m, \quad \frac{d^2 C}{d\phi^2} = 2 (A'_m)^2 + 2 A_m A''_m, \quad (22)$$

$$A'_m = A_m^\phi \alpha_m e^{\alpha_m\phi}, \quad A''_m = A_m^\phi \alpha_m^2 e^{\alpha_m\phi}. \quad (23)$$

All existing Q_{scf} , δQ_{scf} (B-coefficient) machinery for the qcdm background and perturbations is reused unchanged.

Matter perturbations (check: must match 2212.08098 and repo papers)

With the identification $d \ln m_{\text{DM}}/d\phi \leftrightarrow \Lambda_m$, the coupled DM perturbation equations in synchronous gauge are exactly Eqs. (15)–(17) of 2212.08098 (full density contrast δ_c , $\dot{} = d/d\tau$ there):

$$\dot{\delta}_c + \theta_c + \frac{\dot{h}}{2} = \Lambda_m \dot{\delta}\phi + \Lambda_{2,m} \dot{\phi} \delta\phi, \quad (24)$$

$$\dot{\theta}_c + aH\theta_c = \Lambda_m k^2 \delta\phi - \Lambda_m \dot{\phi} \theta_c, \quad (25)$$

$$\ddot{\delta}\phi + 2aH \dot{\delta}\phi + \left(k^2 + a^2 \frac{d^2 V}{d\phi^2}\right) \delta\phi + \frac{1}{2} \dot{h} \dot{\phi} = -a^2 \rho_c [\Lambda_m \delta_c + \Lambda_{2,m} \delta\phi] \quad (\text{physical units; } \times 3 \text{ for } \rho_{\text{class}}). \quad (26)$$

These are what the existing qcdm δQ machinery must reproduce in the pure conformal limit (cross-check of the code, not new code). The genuinely new content is the radiation sector below.

5 DR–DE coupling: equations to implement (new idr_de switch)

Background (no thermodynamics changes; $\sigma, l \geq 2$ hierarchy untouched):

$$\rho_{\text{idr}} = A_r(\phi) \bar{\rho}_{\text{idr}}, \quad \bar{\rho}_{\text{idr}} \propto a^{-4}, \quad p_{\text{idr}} = \frac{1}{3} \rho_{\text{idr}}, \quad (27)$$

$$\phi'' + 2\mathcal{H}\phi' + a^2 \left(V_{,\phi} + \underbrace{3 \bar{\rho}_{\text{qcdm}} A'_m}_{\text{existing } (-Q)} + \underbrace{3 \bar{\rho}_{\text{idr}} A'_r}_{\text{new}} \right) = 0 \quad (\text{CLASS units}). \quad (28)$$

Energy transfer: $\dot{\rho}_{\text{idr}} + 4H\rho_{\text{idr}} = (A'_r/A_r) \dot{\phi} \rho_{\text{idr}}$.

Perturbations, written for the *full* (coupled) variables $\delta_{\text{idr}} \equiv \delta\rho_{\text{idr}}/\rho_{\text{idr}}$ and θ_{idr} , synchronous gauge, with $\Lambda_r \equiv A'_r/A_r$, $\Lambda_{2,r} \equiv A''_r/A_r - \Lambda_r^2$:

$$\delta'_{\text{idr}} = -\frac{4}{3} \left(\theta_{\text{idr}} + \frac{h'}{2} \right) + \Lambda_r \delta\phi' + \Lambda_{2,r} \phi' \delta\phi \quad (29)$$

$$\theta'_{\text{idr}} = \frac{k^2}{4} \delta_{\text{idr}} - k^2 \sigma_{\text{idr}} + \frac{3}{4} \Lambda_r k^2 \delta\phi - \Lambda_r \phi' \theta_{\text{idr}} \quad (30)$$

Derivation note for (22): the r -sector obeys $\nabla_\mu T^\mu{}_\nu(r) = -A'_r \bar{\rho}_r \partial_\nu \phi$ (from the ϕ equation of motion and total conservation; the board's "new term" in (10) is the $A'_r \delta\phi$ piece of $\partial_j \delta p_r$, internal to the r -sector stress divergence). Momentum conservation with $\rho_r + p_r = \frac{4}{3}\rho_r$ and $\rho'_r = -4\mathcal{H}\rho_r + \Lambda_r \phi' \rho_r$ then gives (22). Consistency check in bar variables ($\delta_{\text{idr}} = \bar{\delta} + \Lambda_r \delta\phi$, $\theta_{\text{idr}} = \bar{\theta}$):

$$\bar{\delta}' = -\frac{4}{3}(\bar{\theta} + \frac{h'}{2}), \quad \bar{\theta}' = \frac{k^2}{4} \bar{\delta} - k^2 \bar{\sigma} + \Lambda_r k^2 \delta\phi - \Lambda_r \phi' \bar{\theta}, \quad (31)$$

i.e. the fifth-force term $+\Lambda_r k^2 \delta\phi$ has *the same coefficient as the matter equation* (17), as it must (same covariant force structure per unit inertia). [v1 of this note had $-\frac{k^2}{4} \Lambda_r \delta\phi$ in (22): that was wrong, fixed against 2212.08098 Eq. (17).]

Perturbed scalar field equation: same structure as (18) with the new radiation source added (both sectors written in full- δ variables):

$$\delta\phi'' + 2\mathcal{H} \delta\phi' + (k^2 + a^2 V_{,\phi\phi}) \delta\phi + \frac{1}{2} h' \phi' = -a^2 \sum_{s=c, \text{idr}} \rho_s [\Lambda_s \delta_s + \Lambda_{2,s} \delta\phi] \quad (32)$$

(physical units; $\rho_{\text{phys}} = 3\rho_{\text{class}}$; the $s = c$ term is the existing δQ machinery, the $s = \text{idr}$ term is new). Equivalent bar form of the source: $-a^2[A'_r \bar{\rho}_{\text{idr}} \delta\phi + A'_r \bar{\rho}_{\text{idr}} \bar{\delta}_{\text{idr}}]$ with $\bar{\delta}_{\text{idr}} = \delta_{\text{idr}} - \Lambda_r \delta\phi$.

6 Open choices / questions

1. **Normalization.** Eq. (4) gives $A(0) = 1$, while the paper normalizes $A(\phi_0) = 1$ *today*. Options for the idr sector: (a) shooting so that $A_r(\phi_0) = 1$ and Ω_{idr}^0 is the true density today (analog of `scf_shoot_C0`); (b) let Ω_{idr}^0 parametrize $\bar{\rho}_{\text{idr}}$ only. Same question for the matter sector via $C(\phi_0)$. [choose]
2. **Initial conditions.** Adiabatic ICs for idr kept standard, assuming $A'_r \phi' \rightarrow$ negligible at early times (thawing, ϕ frozen). [confirm]
3. **Euler equation** (22): now anchored to 2212.08098 Eqs. (16)–(17) via the bar-variable consistency check (fifth force $+\Lambda_r k^2 \delta\phi$, drag $-\Lambda_r \phi' \bar{\theta}$, matching the matter sector coefficient-for-coefficient). [re-check the $\frac{3}{4}$ vs 1 coefficient bookkeeping between full and bar variables]